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Investigating the Temperature-Dependent Kinetics in Humidity-Resilient Tin–Titanium-Based Metal Oxide Gas Sensors

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Abstract: Humidity is a well-known interference factor in metal oxide (MOX) gas sensors, significantly impacting their performance in various applications such as environmental monitoring and medical diagnostics. This study investigates the effects of adsorbed water on MOX conductivity using two different materials: pure tin oxide (SnO₂) and a tin–titanium–niobium oxide mixture (SnTiNb)_xO₂ (STN). The results reveal that (SnTiNb)_xO₂ sensors exhibit reduced sensitivity to humidity compared to pure tin oxide, rendering them more suitable for applications where humidity presence is critical. We aimed to shed light on a still controversial debate over the mechanisms involved in the water surface interactions for the aforementioned materials also by exploring theoretical studies in the literature. Experimental analysis involves varying temperatures (100 to 800 °C) to understand the kinetics of surface reactions. Additionally, a brief high-temperature heating method is demonstrated to effectively remove adsorbed humidity from sensor surfaces. The study employs Arrhenius-like plots for graphical interpretation, providing insights into various water adsorption/desorption phenomena. Overall, this research contributes to a deeper understanding of the role of humidity in MOX gas sensor mechanisms and offers practical insights for sensor design and optimization.

Keywords: physisorption; chemisorption; water interactions; gas sensors; chemoresistivity; metal oxide; nanostructures; water vapor; Arrhenius plot; low-temperature water adsorption; tin–titanium oxides



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1. Introduction

It is well known that humidity is a very common interfering species for many gas sensors, in particular for metal oxide (MOX) ones [1,2]. Since humidity is an omnipresent parameter in metal oxide gas sensor applications, e.g., environmental monitoring, medical diagnostics [3–6], and agriculture [7], the study of its effects on gas sensor conduction mechanism is crucial. The water vapor contained in air adsorbs unavoidably onto an oxide surface unless the latter is heated at a high temperature [8]. Nevertheless, water vapor could strongly affect the oxide surface, resulting a serious problem if this oxide is used in the gas sensing field. A hydroxylated surface takes place when OH[−] and H⁺ ions are attracted, respectively, by cations (metal) and anions (oxygen) of the metal oxide lattice, and chemisorbed onto its surface. These sites, performing a high electrostatic attraction on ions, are neutralized by chemisorbed water. On the other hand, due to the weaker attraction, physisorbed water is desorbed at relatively low temperatures. The water presence modulates the electron interactions to (and from) the metal oxide semiconductor, reducing the adsorption of oxygen (as O₂[−] or O[−]) species. Both these phenomena contribute

to a material conductivity enhancement in air but often suppressing the sensor sensitivity to hydrocarbons [9]. Considering the relevance and the unavoidability of the water vapor effects, it is crucial to understand its role in sensing mechanisms. Here, the adsorbed water effects on metal oxide conductivity are investigated employing two different MOX materials: pure tin oxide (SnO_2) and a mixture of tin, titanium, and niobium oxide ($(\text{SnTiNb})_x\text{O}_2$ (STN) with an atomic ratio of $\text{Sn}:\text{Ti}:\text{Nb} = 100:30:5$.

While tin oxide and titanium oxide share a similar crystal structure, their interaction with water vapor differs significantly. Studies suggest that heat is not the sole contributor to water dissociation on the metal oxide surface. Chemisorption and physisorption can be preferably promoted by one or the other material due to various mechanisms:

- Unit cell size: The SnO_2 larger unit cell reduces the influence of hydrogen bonding between adsorbed water molecules compared to TiO_2 [10].
- Bond character: The higher covalent character of $\text{Sn}-\text{O}$ bonds suggests a stronger Brønsted acidity compared to $\text{Ti}-\text{O}$ bonds, potentially promoting dissociation on SnO_2 [10];
- Water adsorption layers: The structure of the first layer of adsorbed water molecules dictates subsequent layers. TiO_2 tends towards associative adsorption (intact molecules), while SnO_2 favors dissociation [11].

These factors are likely to contribute to the observed differences in wet conductivity. SnO_2 exhibits a higher affinity for water dissociation at low to medium coverage, leading to higher water content and potentially explaining its higher conductivity compared to TiO_2 which favors molecular adsorption. Additionally, $\text{Sn}-\text{Ti}$ mixed oxides may exhibit a reduced tendency for dissociation, suggesting a composition-dependent water desorption temperature.

By analyzing these theoretical and new experimental findings, we aim to further elucidate the complex interaction between water vapor and metal oxide surfaces in gas sensors. The results reveal that $(\text{SnTiNb})_x\text{O}_2$ sensors exhibit reduced sensitivity to humidity compared to pure tin oxide, rendering them more suitable for applications where humidity presence is critical.

The crucial role of temperature to understand the conductance kinetics related to sensor surface reactions requires a careful experimental analysis involving hydrated and non-hydrated surfaces, placed in both dry and humid air, by heating them in a wide range of temperatures (100 to 800 °C). Furthermore, the efficacy of a brief high-temperature heating in removing adsorbed humidity from the sensor surface has been demonstrated. This approach promotes the understanding of various water adsorption/desorption phenomena, by using Arrhenius-like (it should be noticed that from this point forward, “Arrhenius *” plot and “Arrhenius-like” plot will be used interchangeably throughout the document to address and abbreviate the more appropriate description for the relationship between conductance and inverse temperature in a “semilog plot” as explained in Appendix A) plots for graphical interpretation and elucidation.

2. Materials and Methods

2.1. SnO_2 , $\text{Sn}_{0.75}\text{Ti}_{0.25}\text{O}_2$, $(\text{SnTi})_x\text{O}_2$ Synthesis

All the reagents were from Sigma Aldrich and were used without any further purification. SnO_2 was synthesized using a simple sol-gel method [12,13]. The typical process involved dissolving 0.01 mol of tin(II) ethylhexanoate in 70 mL of 2-propanol at room temperature, using mechanical stirring. After the reagent was fully dissolved, 30 mL of deionized water was added dropwise. To give the solution an acidic character, which favored the hydrolysis and condensation steps typical of the sol-gel process, 1 mL of 0.1 M HNO_3 was also added. The solution was then stirred for an additional 2 h to complete the formation of the SnO_2 -based gel. Next, the solution was left to settle overnight, allowing the sol and gel phases to separate. The gel was then filtered from the solution and washed with 2-propanol and water to remove byproducts. Finally, the product was dried at 100 °C for 4 h and then calcined at 650 °C for 2 h to obtain SnO_2 nanoparticles.

$\text{Sn}_{0.75}\text{Ti}_{0.25}\text{O}_2$ was synthesized using the co-precipitation method [14,15]. In a typical synthesis, tin(II) ethylhexanoate and titanium(IV) butoxide were dissolved in 70 mL of 2-propanol under stirring. The total molar concentration of $[\text{Ti}^{4+} + \text{Sn}^{2+}]$ was 0.1 mol 0.1 M,

with a Sn:Ti molar ratio of 75:25. Once the reagents were fully dissolved, 30 mL of deionized water was added dropwise to form the SnTiO_2 gel. Similar to the SnO_2 synthesis, 1 mL of HNO_3 was added dropwise to the solution to favor the hydrolysis and condensation reactions, facilitating the co-precipitation of the SnTiO_2 solid solution. The solution was then left to settle overnight, allowing the gel to precipitate. The gel was filtered from the sol and washed several times with deionized water and 2-propanol. Finally, the product was dried at 100 °C for 4 h and then calcined at 650 °C for 2 h to obtain SnTiO_2 nanoparticles.

$(\text{SnTiNb})_x\text{O}_2$ solid solution was synthesized using the co-precipitation method [14,15]. In a typical synthesis, tin(II) ethylhexanoate, titanium(IV) butoxide and niobium(V) chloride were dissolved separately in 2-propanol. Once dissolved, the three solutions were mixed together. The total volume of 2-propanol was 70 mL, the total cation ($\text{Sn}^{2+}\text{Ti}^{4+}\text{Nb}^{5+}$) quantity was 0.01 mol, and the Sn:Ti:Nb ratio was 75:25:5. Afterwards, 30 mL of deionized water was added dropwise to form the $(\text{SnTiNb})_x\text{O}_2$ gel. Similar to the previously described syntheses, 1 mL of HNO_3 was added dropwise to the solution to favor the hydrolysis and condensation reactions. The solution was then left to settle overnight, allowing the gel to precipitate. The gel was filtered from the sol and washed several times with deionized water and 2-propanol. Finally, the product was dried at 100 °C for 4 h and then calcined at 650 °C for 2 h to obtain SnTiO_2 nanoparticles.

2.2. Characterisation and Deposition of Sensing Materials

The morphology and elemental composition of the samples were investigated using a Scanning and Plasma Electron Microscope—Dual Beam CXe Helios 5 PFIB, from ThermoFisher (Thermo Fisher Scientific, Waltham, MA, USA), with an electron beam energy of 15 kV. The system is equipped with a suite that enables high-resolution S/TEM analysis, energy dispersive X-ray spectrometry (EDXS) microanalysis for compositional measurements, and an electron backscatter diffraction (EDSB) detector for crystallographic analysis. Other elemental, morphological, and structural characterizations of the different sensing materials studied in this work are reported elsewhere [14–16]. The SEM-EDX analysis and a summary of the results of previous characterisations performed will be outlined in Section 2.5.

The three different materials were deposited onto alumina substrates (2.54×2.54 mm and ~ 300 μm thick), equipped with an integrated platinum heater (Figure A3 left panel) by means of screen printing. This deposition technique, together with a good control of the MOX paste to be deposited, allows high reproducibility in terms of deposition specifications, including the thickness and porosity of the deposited film [17]. As already reported elsewhere, with the process developed at the Sensors lab of the University of Ferrara, the achievable film thickness is about 30 μm [17–20]. Then, the sensors prepared were bonded to TO39 metal supports (Figure A3 right panel) through thermocompression bonding, enabling their easy integration in the gas chambers for testing [21].

2.3. Research Questions

Often, in scientific research, investigations aimed at a specific purpose unexpectedly lead to discoveries concerning unrelated phenomena. This work represents an in-depth look in the experimental physics field to elucidate how, in some cases, rigorous measurement protocols and clever observations have contributed to advancing the understanding of apparently unrelated phenomena.

As widely discussed in the literature [22], the conductance of metal oxide gas sensors exhibits a repeatable [23] “sigmoid-like” behavior as a function of $1/T$ (already introduced parameter). However, this behavior can significantly change in the presence of different atmospheric gases, especially when exposed to humidity or hydroxylated surfaces [24]; an example of these alterations is shown in Figure 1. Hence, this study was focused on the characterization of the diverse interactions occurring among various materials surface and the adsorbed water molecules, by analyzing the Arrhenius * plot shape variations under different humidity conditions [24].

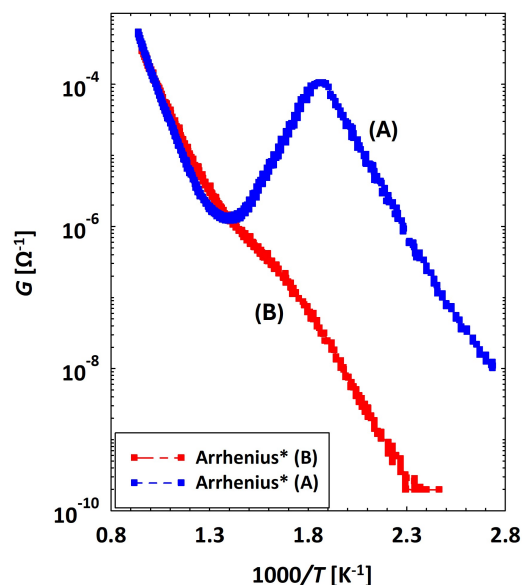


Figure 1. Example of sigmoid-like behavior of conductance (semilog plot y scale) in function of inverse temperature for the same SnO_2 sensor apparently under the same measurement conditions (dry air).

Discrepancy between Conductances of the Same Sensor

Since one Arrhenius plot acquisition takes about two hours ($\sim 3\text{--}5\text{ }^\circ\text{C} \cdot \text{min}^{-1}$ within a $700\text{ }^\circ\text{C}$ range as specified in Appendix A), it is possible to perform at most two/three Arrhenius * plots per day: usually the first was performed in the morning and the second in the afternoon, so leaving to the sensors the proper cooling and recovery time in the middle. Several accurate repeated tests were carried out and showed some important discrepancies (as shown in Figures 2 and 3, taken as examples) which displayed an unexpected, but significant, poor correlation between the shapes of the same sensor Arrhenius * plots, despite being measured sequentially under the same dry air conditions. Throughout this document, the expression “dry air conditions” refers not only to the dry air flux but also to the condition in which the humidity in the chamber was stable and measured (constantly registered, as the sensor signal, every 5 s during the whole measurements) to be nearly 0%, using a commercial *Honeywell HIH-4000* humidity sensor.

Before proceeding, it is important to note that experimental uncertainties will not be displayed on any charts presented in this work. Given their small magnitudes, they would be virtually invisible. For details on the uncertainty magnitudes for G and T , please refer to Appendices B.1 and C.

Figure 2 shows some comparisons of various Arrhenius * plots performed in synthetic dry air flow for two SnO_2 sensors, starting from two different initial conditions: 1st curves (blue in panels (A) and (B)) performed after 12 h at $450\text{ }^\circ\text{C}$ in static environmental air; 2nd (red), after the previous Arrhenius (so after being at above $800\text{ }^\circ\text{C}$ in the last part of the measure) and after 2 h at $450\text{ }^\circ\text{C}$ in continuous dry air flow.

The first Arrhenius curve, for both charts, as shown in Figure 2, begins with a very different (much higher) conductance value across a wide range of temperatures, ending up overlapping in the last part of the curve (for high temperatures), where it adopts the very same values of the second one.

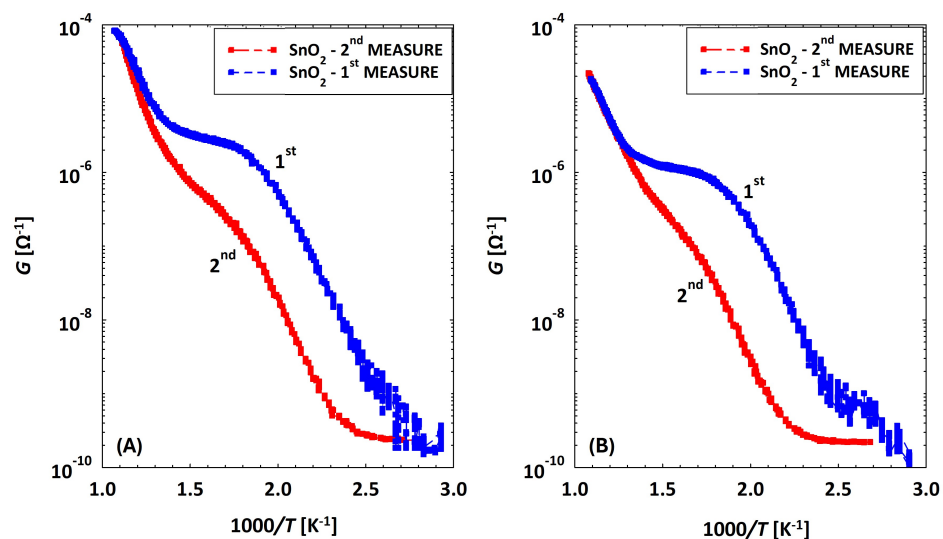


Figure 2. Comparison between Arrhenius-like plots performed in dry air on two different SnO₂ sensors within two different days. The 1st and 2nd mean the first and second of the same day, respectively. The (A,B) panels refer to two different sensors.

These experiments, given the huge number of variables, must be thoroughly performed in the practical part as well as in the data examining and processing; given that, all the same sensor curves of Figure 2, even if very different in shape and values, showed the same behavior in the final part concomitantly with the highest temperatures, where the curves were overlapped. In Figure 3, instead, the discrepancies between curves, i.e., the first and second plots of the same day, are less noticeable, showing their superposition in concomitance of the lowest- and the highest-temperature regions.

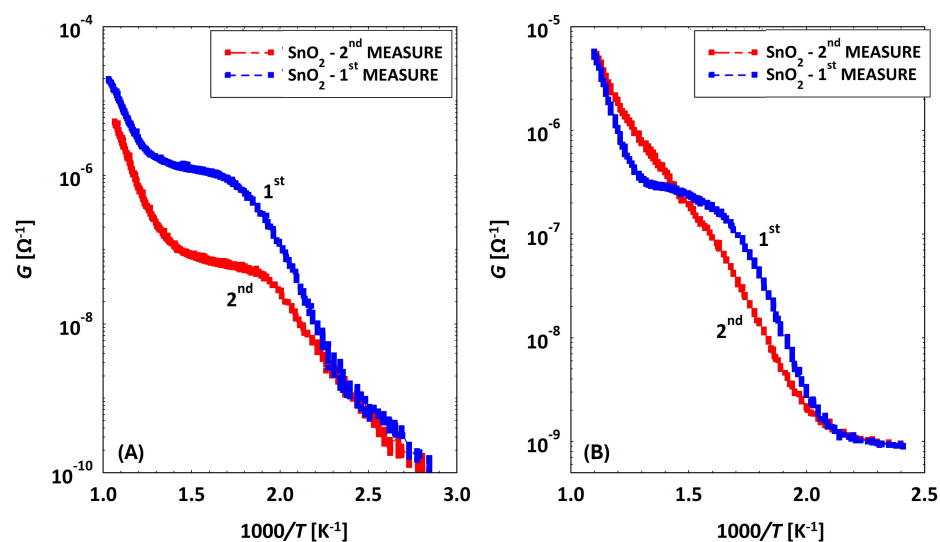


Figure 3. Comparison between Arrhenius-like plots performed in dry air on two different SnO₂ sensors within two different days. The 1st and 2nd mean the first and second of the same day, respectively. The (A,B) panels refer to two different sensors.

At this juncture, it became crucial to comprehend why the Arrhenius-like plots of conductance were not reproducible, particularly within the temperature range of 150–600 °C. Despite the apparent uniformity in measurement parameters, it was imperative to examine potential hidden boundary conditions that might influence the atmospheric conditions within the chamber. Arrhenius * plots were labeled as 1st and 2nd, meaning “the first” and “the second” of the same day, respectively. The first one was preceded by

15 h of dry-air flux absence (“no flux”: the air flow was stopped during the night), while the second one was always performed after enough time from the first one, but without any air flux interruption. Because the test chamber is not hermetically sealed from the outside environment (exhaust pipeline is always opened to the outside), the “no flux” condition consisted in a return of moist ambient air over the sensors.

This phenomenon forced the sensors surface rehydration quantitatively depending also on the sensor operating temperature. The shape of the subsequent first Arrhenius * plot is heavily influenced by the arrangement of water layers and on the type of their interaction with the metal oxide. This assumption is corroborated by the fact that, when a third measurement was performed, the second and third Arrhenius * plot shapes for the same sensors overlapped almost perfectly (Figure 4 left panel). Figure 4, right panel, shows three heating ramps performed on the same SnO_2 sensor, with the first and the second ones performed the same day in a row, and the third one performed first on the following day but previously conditioning the sensor at 700°C for 10 min; this preliminary treatment will be called hereinafter “conditioning”.

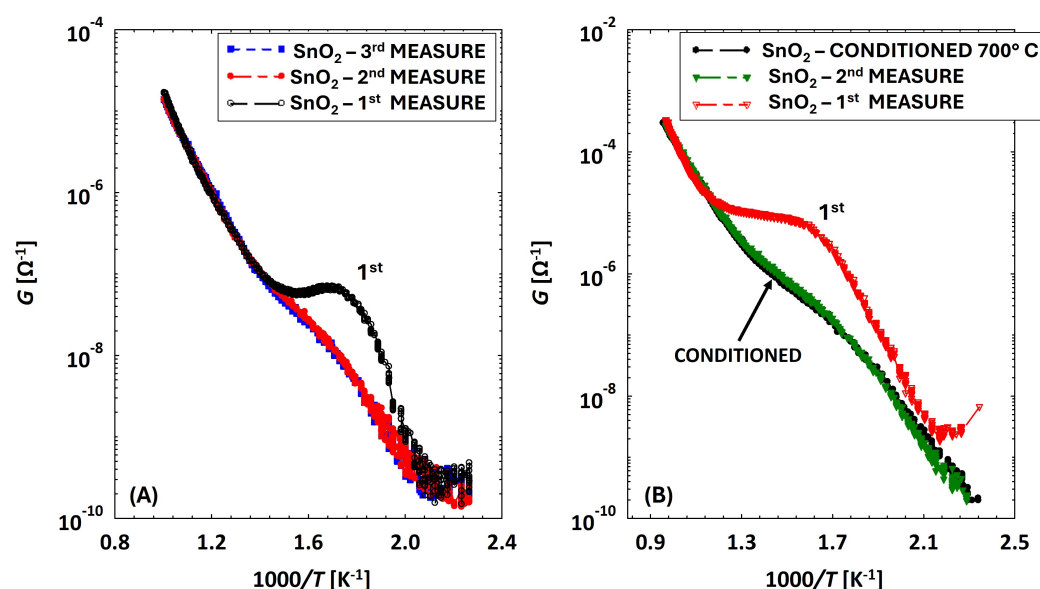


Figure 4. Left panel (A): comparison between three Arrhenius-like plots performed per day on the same SnO_2 . The 1st measure shows a significant different conductance behavior with respect to the 2nd and 3rd on the same day. Right panel (B): 1st measure of the day compared to the 2nd and another one (not in the same day), but in the latter, the sensor was heated at 700°C for 10 min before starting the ramp. It is worth noting that the 2nd and the conditioned curves perfectly superpose.

At present, the irregular (“curly”) shape of the 1st-labeled plots (visible in both the left and right panels of Figure 4) within the x -axis range of $1000/T \sim 2.1\text{--}2.3$ ($T \sim 160\text{--}180^\circ\text{C}$) may seem peculiar, likely due to experimental uncertainties. However, it was precisely this behavior that prompted an exploration into the conductivity trend at lower temperatures than typically examined in this type of analysis. Notably, it was observed that initiating measurements from room temperature ($\sim 25^\circ\text{C}$) might be advisable. Further clarification will be provided in subsequent discussions.

Based on these latest comparisons, it can be noticed that the differences seen before can be definitely attributed to different conditions of each of the sensor experiments at the beginning of the measurement; this is finally confirmed by looking at the four images in Figure 5 that report the inverse temperature dependence of the conductance for four different SnO_2 sensors (similar to most of the previous measurements, the sensors used here are all based on SnO_2 material).

In the upcoming discussion, an in-depth examination of the properties of SnO_2 is described, alongside an investigation of the properties of a specific semiconductor material

named STN. This specific material will be explored as a potential solution to address certain challenges associated with tin dioxide, such as its cross-sensitivity to humidity and other gases.

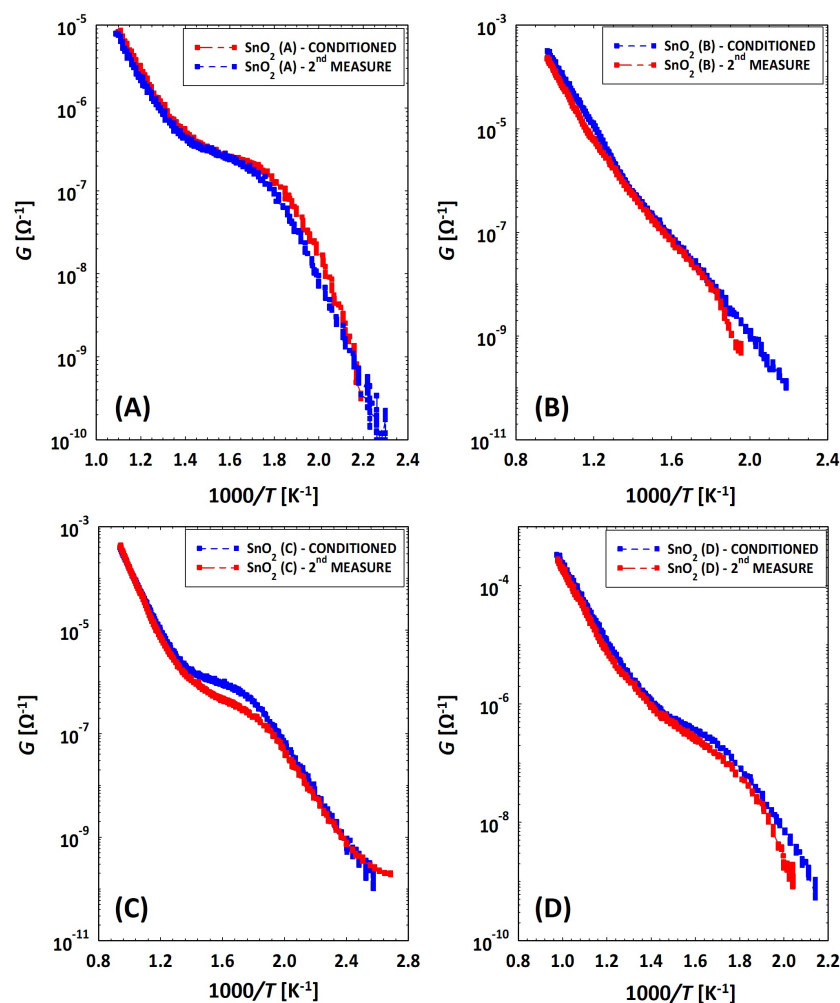


Figure 5. Comparing the second measurement of the day (in red) with a conditioned (at 700 °C for 10 min) one (in blue) for four different SnO₂ sensors (A–D), the *conditioning* process guarantees the same initial conditions for every measure and is essential to ensure the same initial conditions for every measure.

2.4. Humidity Effects on MOX Sensors

As already introduced, given the ubiquity of humidity in real-world gas sensor applications (environmental, medical, and agricultural), studying its impact on sensor conduction is crucial. In this work, the impact of humidity in the conduction mechanisms of two tin oxide-based sensor materials (pure SnO₂ and the STN) has been examined. The results showed that the latter has the feature of being less affected by humidity and is thus beneficial in environmental and medical applications.

It is well known that humidity is a universal interfering gas for most MOX sensors. The adsorption and hydration of water from air onto a MOX is an unavoidable fact, unless the MOX is heated to a high temperature [8]. Moreover, water can have strong effects if the MOX is employed as a gas sensor. As already described, a hydroxylated surface is formed when OH[−] ions are attracted by the oxide cations and vice versa for H⁺ ions that are attracted by anions. Chemisorbed water has the effect of neutralizing those sites that have an intense ion electrostatic attraction. On the other hand, due to the weaker attraction, physisorbed water is eliminated at lower temperatures. When water is present, it modulates the electron interactions to and from the metal oxide semiconductor, weakening the ad-

sorption of oxygen (as O_2^- or O^- species). Both these phenomena contribute to an enhanced conductivity of the material in air; besides, the sensitivity to hydrocarbons is often (even not always) suppressed [25]. A deeper analysis of the water–surface interaction mechanisms is proposed in Appendix D, presenting the state of the art in the theoretical background, whose correlation with the observations is presented throughout the experimental sections.

Experiments began with the analysis of hydrated and non-hydrated surface conductance in dry air versus humid air, passing through a whole range of working temperatures (from 100 to 700 °C), showing how a short high temperature heating can remove the adsorbed humidity.

All along the experimental section, the temperature covers a pivotal role in understanding the kinetics of conductance related to surface reactions. It is worth noting that to maintain simplicity and enhance immediate comprehension, when discussing significant data points on these graphs, temperatures are referenced in degree Celsius (°C). However, it is important to emphasize that throughout the experimental work and calculations, temperatures were consistently recorded and are represented as T in Kelvin (K) and $1/T$ (in K^{-1}) on the x -axis of the Arrhenius * plots.

2.5. Morphological and Chemical Characterization of the Nanomaterial Used

The SEM-EDXS characterization was performed on the powder synthesized for analyzing nanoparticle morphologies and elemental composition. As can be seen from Figure 6, the morphology of all the three different nanomaterials was spherical-like, with an average size of less than 100 nm. EDX spectrometry (insets of Section 2.2) showed the presence of the expected elements in the three different samples, with the addition of a small concentration of carbon probably due to contamination of the sample by exposure to ambient air. The atomic ratio between Sn and Ti in the $SnTiO_2$ powder was 76.3:23.7, which closely matches the target ratio of 75:25. Concerning $(SnTiNb)_xO_2$ sample, the Sn:Ti:Nb ratio was 74.2:26.4:4.4, also in this case very close to the target ratio. Further characterizations performed on these materials and reported in our previous work [14–16] included X-ray powder diffraction (XRD) and X-ray photoelectron spectroscopy (XPS). In particular, the XRPD analysis revealed that the powders were mainly composed of a tetragonal rutile-type phase (space group $P4_2/mnm$), with the presence of tetragonal anatase-type phase (s.g. $I4_1/amd$). The substitution of Sn^{4+} ions in the lattice with Ti^{4+} and Nb^{5+} ions resulted in a change in the unit-cell parameters of $(SnTi)_xO_2$ and $(SnTiNb)_xO_2$ compared to SnO_2 , leading to a decrease in the unit-cell volume due to the smaller ionic radius of Ti^{4+} and Nb^{5+} than Sn^{4+} [15]. XPS characterization allowed to evaluate the surface concentration of the different metal atoms contained in the three samples, which resulted to be higher than the bulk concentration identified using other techniques (e.g., EDX), probably due to a partial surface segregation effect during the nanoparticle synthesis process and the subsequent powder calcination [14,15]. Furthermore, the substitution of Sn with Ti and Nb ions on the lattice resulted in an increase in surface defects for $(SnTi)_xO_2$ and $(SnTiNb)_xO_2$ compared to SnO_2 .

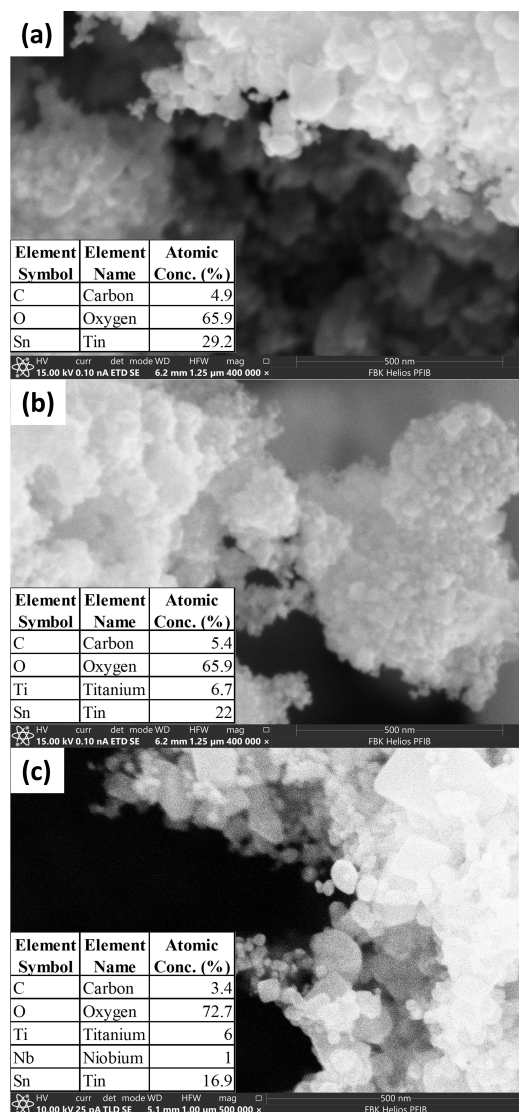


Figure 6. SEM images of (a) SnO_2 , (b) SnTiO_2 , and (c) $(\text{SnTiNb})_x\text{O}_2$ powders. The elemental composition of the three samples, obtained through EDXS analysis, are reported as insets.

2.6. Gas Test Bench Setup

The sensors behavior was studied under controlled conditions of temperature, air-flow and gas concentration by housing them in a hermetically sealed chamber of about 250 cm^3 (Figure 7) made of aluminum that also hosts temperature and humidity sensors. The technical gases from the bottles were uniformly diffused from the chamber base. Gas concentrations were calibrated thanks to mass-flow controllers and a specific software and conveyed to the chamber by means of Teflon pipelines with a constant dynamic flow of 500 sccm (standard cubic centimeter per minute).

Chemisorbed species on semiconductor layers are responsible for trapping and/or transferring electronic carriers to the underlying oxide, hence changing its electrical characteristics. In particular, chemisorbed oxygen in the form of anions (O_2^- , O^- , and O^{2-}) plays an important role in the sensing mechanism by altering interactions with target gases. These gases are generally reducing agents, like hydrocarbons or, particularly, CO.

The Arrhenius plot (or, similarly, the conductivity trend in the function of the inverse temperature; see Appendix A) is one of the most significant techniques to study semiconductor oxide sensors behavior in air; as such, it has been used in this study to comprehend the interactions occurring by adsorbed/chemisorbed oxygen and water species.

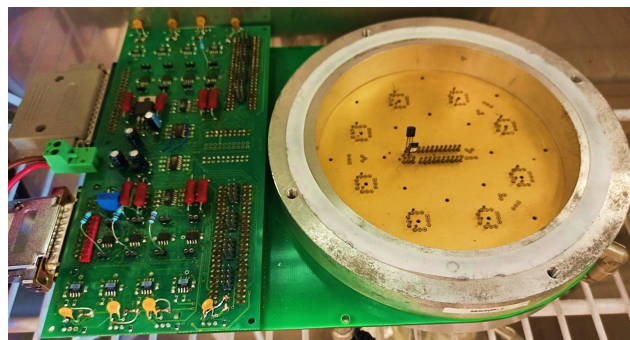


Figure 7. Laboratory chamber for gas tests. It can hold eight sensors; air and gases flow from the bottom inlet to top exhaust outlet.

2.7. SnO_2

Repeated Arrhenius-like plots have been performed sequentially and after different preliminary heating of sensors in dry air, aiming to desorb humidity and to compare them among each other.

An explanation of what is described in Section 2.3 for Figure 2 must be attributed to water molecules interacting (physisorption/chemisorption) with the sensor's surface during the overnight leakage in the chamber (condition 1). These water molecules increase the sensor's conductivity by acting like *donors* (see Appendix D for details).

A distinct scenario emerged when Arrhenius plots were generated after complete surface dehydration; here, it is highly probable that the predominant interactive species that modulates the conductivity were various oxygen forms (O_2^- , O^- , and eventually O^{2-}) [25]. To show the repeatability of this phenomenon, both the measurements of Figure 2 were replicated in two different days as Figure 8 displays.

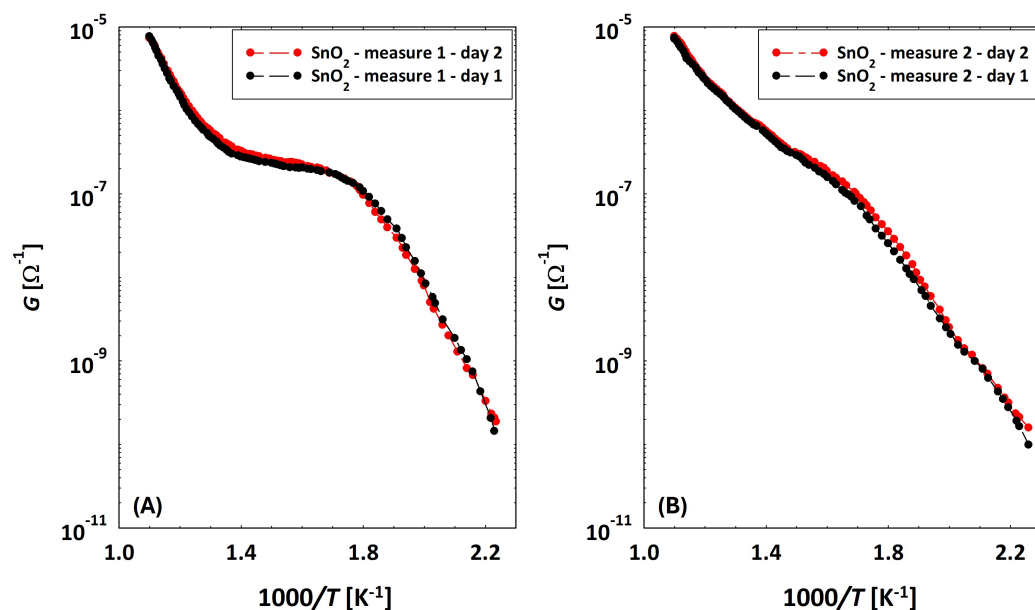


Figure 8. Arrhenius * plots of a tin dioxide (SnO_2) sensor measured on two separate days: panels (A,B). Both measurements used the same initial conditions, resulting in identical curves. Measure 1 (left panel): the first measurement of the day, performed after the sensor spent the night rehydrated in still air (no gas flow). Measure 2 (right panel): performed immediately after Measure 1 on both days. Despite starting with very different values and shapes, conductances in measure 1 and measure 2, overlap above $\sim 650^\circ\text{C}$.

From the measurements shown in Figures 2–4 and 8, it is clear that at temperatures of about 550 to 600°C , the conductance curves start to overlap. To gain a deeper understand-

ing of this phenomenon, multiple conductance measurements on the same sensor across a wide range of temperatures were conducted.

As Figure 9 illustrates, the initial amount of surface hydroxylation significantly impacts the resulting Arrhenius * plot. Here, all curves correspond to the same SnO₂ sensor, with efforts made to ensure slightly different initial surface water content [24]. For measurement (A), the sensor was used for the first time without prior heating in dry air. The exposure of the sensor surface directly to ambient air for a long time led to the adsorption of a significant amount of humidity. In (B), (C), and (D), the sensor was left in the measurement chamber without dry air flow overnight; therefore, less humidity was able to enter the measurement chamber.

The different initial heating conditions completed the scenario gradually decreasing the amount of humidity on the surface, this reflecting in the progressive flattening of the curves. In the initial sections of conductivity curves (C) and (D) (from 2.3 to 1.8 going right to left on the abscissa, in the temperature range of 150 °C to about 280–320 °C), it is noteworthy that their conductance overlaps. This observation, when considered in the context of the various water adsorption phenomena detailed in Appendix D, suggests that most of the physisorbed water molecules had already desorbed from the surface due to the preheating at 450 °C.

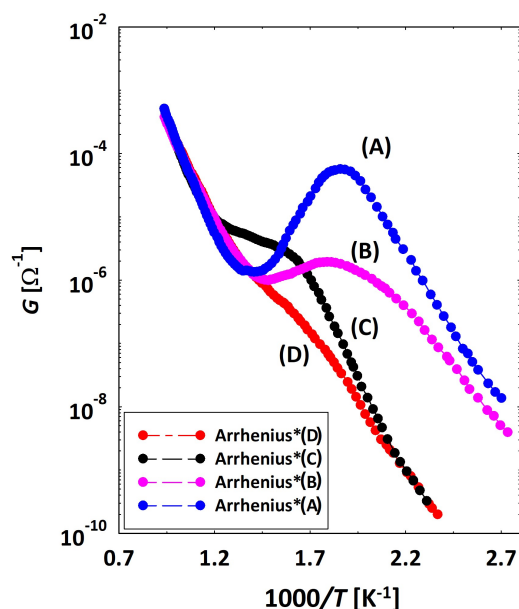


Figure 9. Arrhenius plots carried out on a SnO₂ sensor under different conditions: Condition A: the sensor was first heated to 100 °C and kept in dry air for 10 min before measurements. Condition B: the sensor underwent the same treatment as condition A but was then left at room temperature for 24 h before measurements. Condition C: the sensor was heated to a higher temperature, 450 °C, and kept in dry air for 10 min before measurements. Condition D: the sensor was heated to an even higher temperature, 700 °C, and kept in dry air for 10 min before measurements.

Hydrogen-bonded H₂O molecules are known to be thermally unstable and easily desorb at low temperatures. Consequently, it is likely that the surface contains less than a water monolayer at this point, precluding hydronium transport and proton hopping conduction mechanisms at lower temperatures.

Furthermore, it becomes evident that incompletely removed water molecules require additional heat to initiate the chemisorption process (that leads to electron injection in the conduction band) and even more heat for complete desorption. This phenomenon is reflected in the increase in conductance occurring in the intermediate portion of curve (C), prior to the eventual desorption observed in the latter part of the curve.

This seems to confirm that at low temperatures and in high relative humidity conditions, the conduction is dominated [26,27] by the proton and hydronium hopping mechanisms. It can be also guessed that the less noticeable inflection in the middle part of the curve (D) may be due to the solely oxygen species modification from O_2^- to O^- on the surface of the metal oxide because the preventive heating at 700 °C has provoked the desorption of all the water species.

To confirm what has been described so far, Arrhenius * plots were performed and compared with each other, one after having performed another Arrhenius * before them, and one after having heated sensors at 700 °C, both in synthetic dry airflow for 10 min and both starting after 12 h in environmental air and room temperature. Comparison between those is shown in Figure 10 for two different SnO_2 sensors (as well as in the already shown Figure 5). This proved that the sensor surfaces result completely dehydrated after the conditioning process.

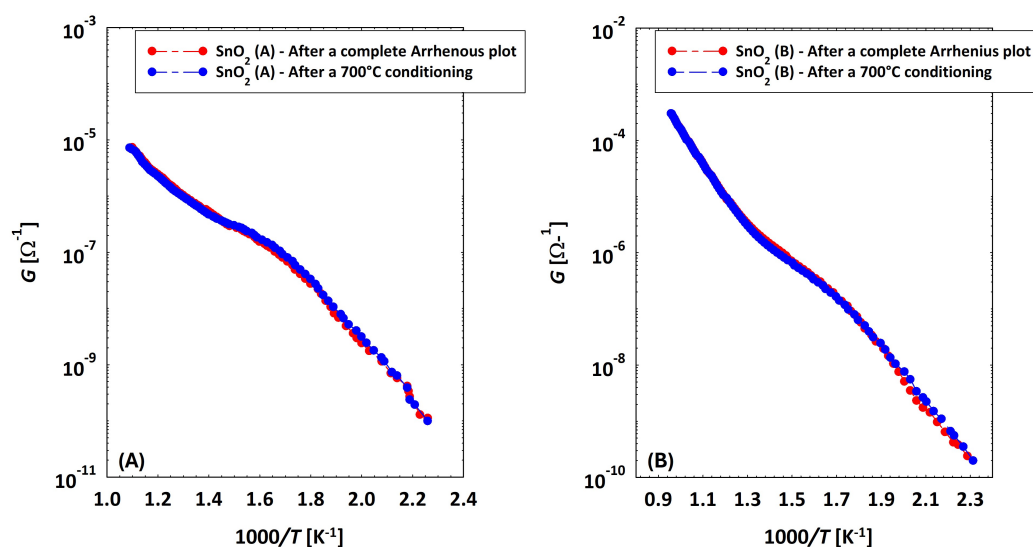


Figure 10. Comparison between Arrhenius plots for two different SnO_2 sensors, (A,B), after a first Arrhenius and after heating at 700 °C. Both measurements were taken in synthetic dry air for 10 min and both starting after 12 h in environmental air and at room temperature.

Summarizing, the surface barrier and, consequently, the temperature stimulated conductance, is strongly affected by adsorbed/chemisorbed water species on the sensor surface: the measurement are performed starting from a temperature of about 90–120 °C (around the value 2.5 on the $1000/T$ axis), and from this temperature up to about 250–300 °C there is a greater slope on the first part on the curves in humid air conditions, which denotes an increasing conductivity in a faster way. At this point, the slope of the curve begins to bend, and this is attributed to the fact that the O_2^- species initially adsorbed on the material begin to transform into O^- , and at higher temperatures, into O^{2-} species, by absorbing energy supplied by heating.

This phase of the process is the one that presents the maximum surface barrier since the oxygen ions extract the highest number of electrons from the semiconductor nanoparticles, becoming O^{2-} and increasing the thickness of the depletion layer in the nanoparticles themselves. In the presence of humidity, physisorbed water starts to be desorbed at lower temperatures but also starts to be chemisorbed in the form of ions OH^- and H^+ ; the result is a widening of the temperature range between the first and the second slope changes, right where the lowering of the potential barrier height takes place (O^- and OH^- adsorption), given that the two processes are concurrent, because the presence of water vapor has an effect similar to that of reducing species which are adsorbed on the sensitive material, increasing its conductivity.

When the phenomenon of the transformation of the oxygen species is finished, the conductivity starts to grow again with the temperature; in the case of humid air conditions, this restoration of conductivity growth takes place at a higher temperature since the transformation from O_2^- to O^- species is thwarted by the presence of the species OH^- [22].

It is indeed intriguing and instructive to examine Arrhenius * plots where the conductance, as a function of the inverse temperature of an *unconditioned* sensor when measured in a dry air environment, exhibits variations from those observed in a conditioned sensor, primarily in specific regions of the chart.

Referring to Figure 11, those regions are located in the low-, medium-, and high-temperature ranges, here numbered with 1, 2, and 3.

As a starting point, considering that the red curve corresponds to the non-conditioned case, it is reasonable to assume a certain level of humidity on the sensor surface. However, in region 1 (temperatures between 160 °C and 200 °C), the two curves overlap, reminiscent of the behavior observed between curves (C) and (D) in Figure 9, as well as in the right panel of Figure 3. This phenomenon has previously been attributed to the presence of less than a monolayer of physisorbed water molecules. These non-dissociated water molecules appear to have no discernible macroscopic influence on the conduction mechanism in this low-temperature range. In region 2 (temperatures between 200 °C and 400 °C), the two curves begin to diverge.

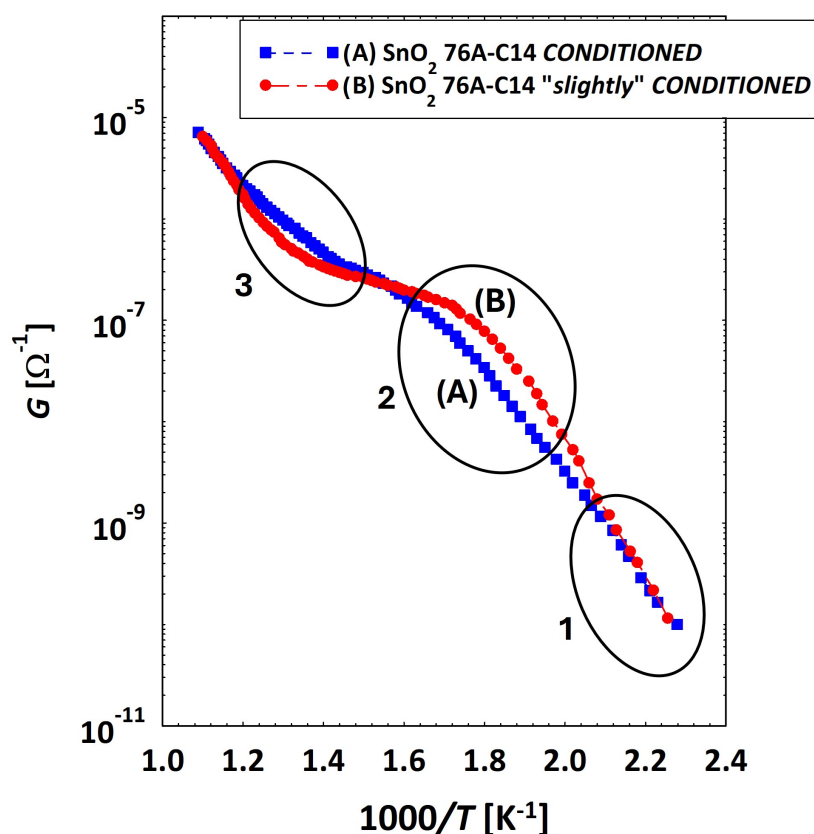


Figure 11. Conductance as function of inverse temperature: incomplete physisorbed water layer ("slightly" conditioned) (B) vs. dehydroxylated surface (A) shapes for the same SnO_2 sensor.

The progressively increasing conductance of the curve (B) can be attributed to the growing number of chemisorbed water molecules, which play a pivotal role in electron injection into the conduction band. Due to the relatively subtle difference between the two curves, the curve (B) in the chart legend is denoted as "*slightly conditioned*". This designation underscores the presence of a reduced quantity of water on the sensor surface. This

configuration, while not uncommon, is intricately linked to specific initial and boundary conditions that are not always finely controllable, making it less readily attainable.

In the third and final step (region 3), which occurs in the temperature range of 400 to 600 °C, an interesting phenomenon comes into play. During this phase, chemisorbed water molecules undergo a desorption process driven by thermal energy. In some instances, like the one observed here, the conductance exhibits a more pronounced inflection compared to the measurements taken under completely dry conditions.

This behavior suggests that, in such cases, hydroxyl OH^- species do not desorb as a whole entity [22]. Instead, hydrogen atoms tend to leave the surface first, leading to an excess of O^- and O^{2-} species. These oxygen-based extra species appear to contribute significantly to the increase in resistivity in this specific temperature range.

2.8. SnTiNbO_2 —(STN)

In our previous work ([24]), we investigated the different behavior of SnO_2 and the STN materials under the impact of water interaction with the surface. Starting from there, this and the next sections aim to provide a more comprehensive theoretical framework and an improved experimental effort by incorporating recent advancements in this specific topic. This, along with the idea to investigate the water–surface interactions at lower than usual temperatures, will allow for a deeper understanding of the underlying mechanisms at play. What emerges is that the mixed Sn–Ti materials experiment shows very little interaction with humidity, showing good overlap between dry and wet plots.

Experiments showed that there is a significantly reduced discrepancy between Arrhenius * plot shapes performed one after the other or after preliminary conditioning in dry or wet air flux. In particular, the temperature at which the sensor is preheated (for 10 min) before the Arrhenius ramp notably affects the surface water content, this reflecting (at least in the SnO_2 case) in a very different conductance shape over the temperature range.

In Figure 12, the same initial conditions (1 V, 5 V and 7 V corresponding to about ~100 °C, ~500 °C, ~700 °C) are set for three different STN but in a wet atmosphere (relative humidity of 20%). Also here, there is very good reproducibility between different sensors and between different conditionings.

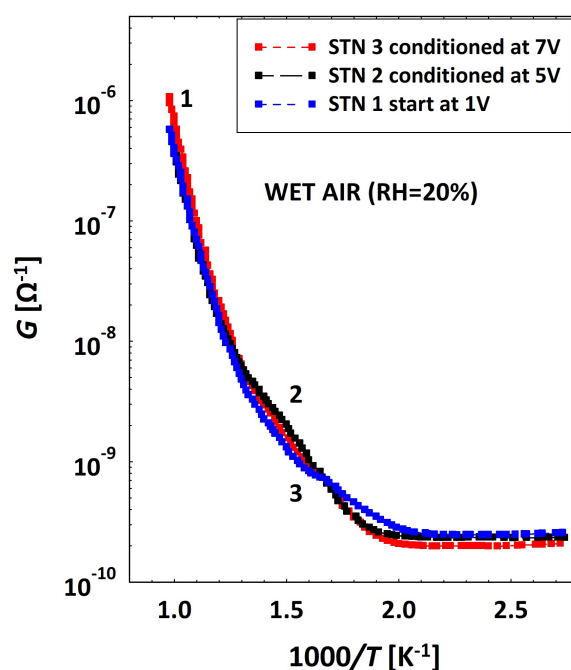


Figure 12. Arrhenius plots in WET air (RH = 20%) comparison between three STN sensors; one with no preliminary heating (starting at 1 V~100 °C). The second conditioned at 5 V~500 °C). The third sensor conditioned at 7 V~700 °C).

Figure 13 shows how similar the conductances of the STN material are when the Arrhenius plot is performed in dry or wet air. But also, a noticeable difference between the two conditions is visible in the first part of the heating process if it starts from a lower than standard temperature (about 50 °C), and from that, up to about 170 °C is the range in which probably the greater part of the water is slowly removed from the surface. This led to the decision to widen the usual Arrhenius plot temperature range from ambient (where the differences are more evident) to approximately 700–800 °C to identify any other potential differences at high temperatures.

It is therefore mandatory to investigate further into the phenomena associated to these distinct behaviors, because it is now evident that these variations cannot be solely attributed to externally provided energy.

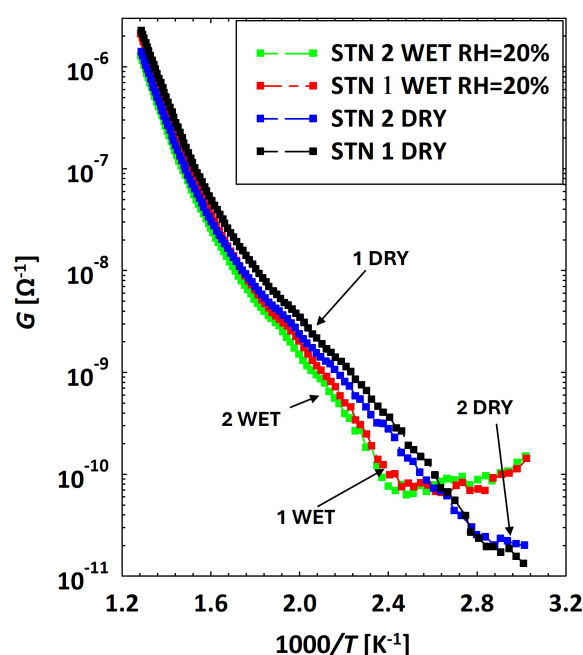


Figure 13. Comparison among repeated dry–wet Arrhenius plots for two different (1 and 2) STN sensors.

3. Results and Discussion

3.1. Theoretical Background for Wet Conductivity in SnO_2 , TiO_2 and $(\text{Sn-Ti})\text{O}_2$ Materials

Other studies highlight a lower conductivity of TiO_2 -based sensors that also have a higher thermodynamic stability and less cross-sensitivity to humidity than SnO_2 [28], this being due to a series of different mechanisms in H_2O adsorption between the two materials [11].

The purpose of this section is to investigate what, other than heat, can or cannot provide the activation energy needed for the dissociation of water molecules over a metal oxide surface.

Given the critical significance of water's interaction with semiconductor-based sensors, extensive research efforts have been directed toward this subject, encompassing both theoretical and experimental investigations [29–31]. However, not all aspects of these interaction mechanisms have been unequivocally elucidated. Controversies persist, particularly concerning the associative/dissociative water adsorption on materials such as SnO_2 , TiO_2 , and their solid solutions as shown in the literature [32,33].

The objective of this section within the current study is to amalgamate various theoretical and experimental findings to add another piece to the puzzle, further advancing the understanding of this intricate area of inquiry.

In a related study, Jerome F. McAleer et al. [22] conducted an investigation into molecular interactions and conductivity on SnO_2 gas sensors. Using frequency relaxation analysis in ac impedance spectroscopy, they found no evidence of the proton hopping

conduction mechanism at low temperatures for this material, concluding that the main interaction of SnO₂ with water must be mainly dissociative.

A combination of different phenomena contributes to this discussion, including geometrical factors. Although both TiO₂ rutile and SnO₂ cassiterite share the same crystal structure, they differ in their unit cell parameters. Lindan P. J. D. suggested [10] that the larger unit cell dimensions of SnO₂ ($a = 4.737$, $c = 3.186$ Å) compared to those of TiO₂ ($a = 4.593$, $c = 2.959$ Å) [34] reduce the energetic significance of H-bonding among adsorbed H₂O molecules on the SnO₂ surface compared to TiO₂.

Furthermore, Lindan P. J. D. proposed that considering the water adsorption process on the surface as an acid/base interaction, the higher covalent character of Sn–O (electronegativity of Sn: 1.96) bonds in comparison to Ti–O (electronegativity of Ti: 1.54) [35] bonds might be another factor favoring dissociation on the tin oxide surface. This suggests [11] that the hydrated Sn(V) ion is potentially a stronger Brønsted acid than the hydrated Ti(V) ion on the rutile surface.

In a separate study, Andrei V. Bandura et al. employed static plane-wave density functional theory (PW DFT) to state that when more than a monolayer of water molecules is adsorbed on a surface, the structure of the first layer (L1) predetermines the structure of the next layer (L2). Consequently, the nature of L1, which is predominantly associative for TiO₂ and dissociative for SnO₂, contributes to the distinction between water hydration on rutile and cassiterite surfaces. This suggests that neither totally molecular nor entirely dissociated species form monolayers on one surface compared to another. Instead, there is a significant presence of molecular water on TiO₂ and a substantial level of dissociation on SnO₂ [11].

Furthermore, Konstanze R. Hahn et al. [28], again employing theoretical calculations based on density functional theory (DFT), shed light on the strong relationship between water adsorption mechanisms and the concentration of Ti in (Sn–Ti)O₂ solid solutions. Notably, they identified a point of minimum binding energies for water adsorption at various coverages, which occurs within the range of 25 to 33% Ti content.

For pure SnO₂, their research revealed a pronounced tendency for water molecules to dissociate at low to medium coverages, with a gradual increase in the presence of molecular H₂O as coverage increases. Moreover, tin oxide exhibits a higher affinity for moisture adsorption than titanium oxide, resulting in an increased water content on its surface.

Additionally, their findings indicated that Sn–Ti mixed solid solutions exhibit a reduced tendency for dissociative interaction with water molecules, and they calculated a global minimum of water content for a Ti content of 25%. This suggests that the temperature required to desorb water molecules from the surface is at its lowest in this composition [28].

3.2. Low Temperatures' Experimental Behavior

When examining Figure 14, we can observe that, at moderate humidity levels (specifically RH = 20%), the conductance of the wet Arrhenius curve begins to overlap with the dry curve at around 180 °C. Prior to this point, the wet conductance is slightly higher, suggesting that at lower temperatures, water predominantly exists in its molecular form, conducting through the Grotthuss mechanism. Beyond 200 °C, it becomes apparent that the water is almost entirely removed.

In curve (C), which corresponds to RH = 40%, there is a transition between proton hopping as the dominant conduction mechanism and the injection of electrons into the conduction band due to the chemisorption process of water. This transition occurs at approximately 110 °C, where the conductance remains higher. This higher conductance persists until complete water desorption, which happens around 300 °C.

The proton hopping/hydronium transport conduction mechanisms gain further support when examining Figure 15, which presents another dry–wet comparison. In this case, the wet curve measurement was initiated at approximately 20 °C. The significant conductivity observed at this temperature is unlikely to be attributed to chemisorbed hydroxyl species since chemisorption typically requires a higher energy input. Again, it can be inferred that water molecules are gradually removed from the surface until those

remaining begin to undergo dissociation, and chemisorption, which here is occurring at around 120 °C, becomes evident. This is further supported by the fact that each inflection point in an Arrhenius plot may correspond to a change in the rate-determining step or a transition between different reaction mechanisms.

So to emphasize the point, the relative minimum at about 120 °C is likely to correspond to a change in activation energy between physisorption and chemisorption for water molecules.

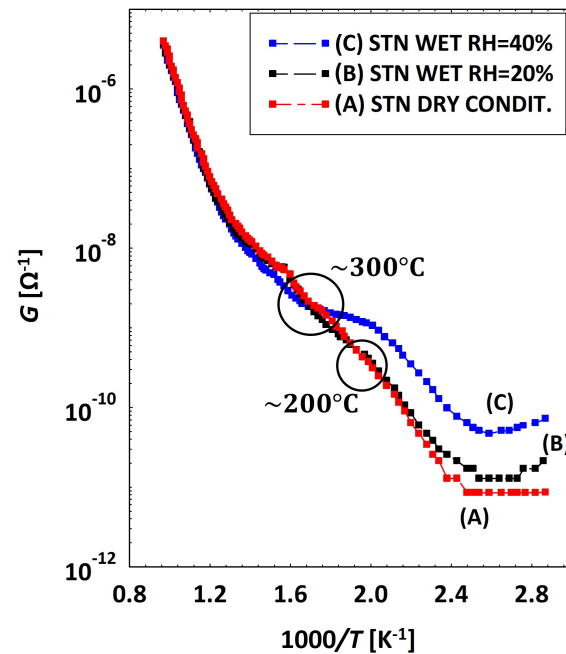


Figure 14. Comparison among: dry and two wet (RH = 20% and RH = 40%) Arrhenius plots for the same STN sensor.

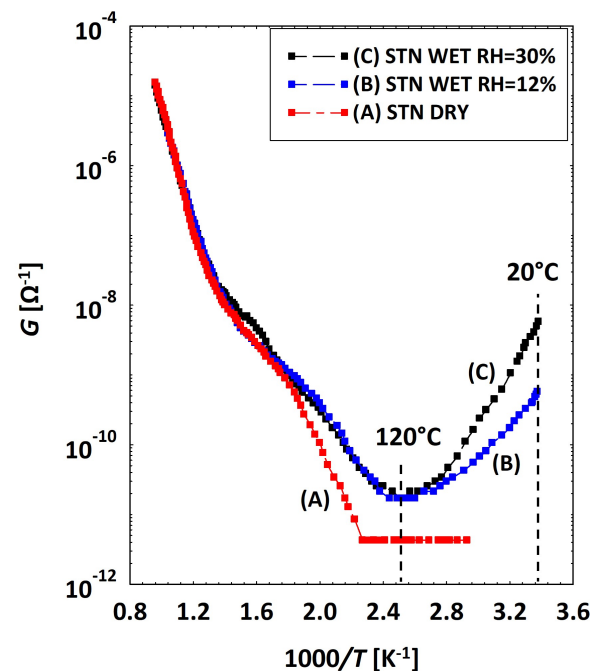


Figure 15. Comparison between a dry air Arrhenius plot and at two humidity levels (RH = 12 and 30%). The wet measure conductivity was started from 20 °C to show how the conductivity shape changes in the low temperatures range. This behavior can be attributed to a non-dissociative water molecules adsorption, giving rise to Grotthuss and hydronium transport mechanisms.

The behavior of the SnO_2 material in a humid environment at low temperatures differs significantly from that observed so far with the STN material. To highlight this difference, Figure 15 can be compared with Figure 16, in which the same SnO_2 sensor is measured in two different wet conditions.

It can be noticed that for humidity at 40% the conductance at low T greatly differs from that at 12%, meaning that there must be a lot of sites at disposal for water molecules to be chemisorbed. Moreover, also in this case, there is evidence of water molecules physisorption at very low temperatures, but the entity of the phenomenon is far less incisive if compared with the STN material. In particular, the temperature at which molecular water starts the chemisorption is less an half compared to STN, meaning that the required activation energy is enhanced by the differences in geometry, electron affinity, and acid/base strength between the two materials as explained by the theoretical insights proposed in Section 3.1.

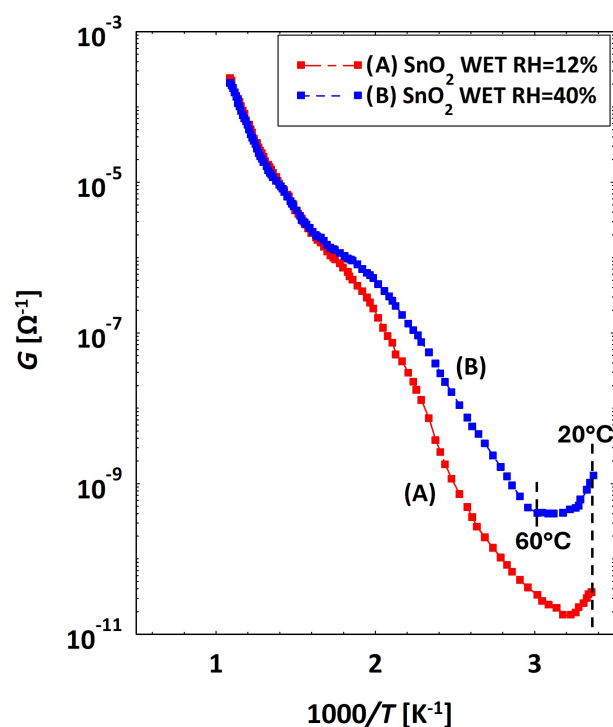


Figure 16. Conductance G vs. inverse T for a SnO_2 sensor. Both measures are conducted in a different percentage of wet air. A low affinity to molecular water physisorption can be inferred by the behavior of this material in the range 20–60 °C

It is indeed enlightening to delve deeper into the low-temperature interaction of the tin oxide material with water molecules. In Figure 17B, the wet atmosphere (40%) conductance behavior is compared to the conductance of the same sensor measured in dry conditions after being exposed at ambient air and at room temperature overnight. Before starting the Arrhenius plot, the sensor is in a flow of dry air and at ~ 27 °C for two hours, and its conductance behavior is reported in Figure 17A, along with the chamber's relative humidity content. What becomes evident is that as conductance decreases, humidity gradually evacuates from the chamber. This observation, coupled with the findings in panel B), once again reaffirms the sensor's capacity to readily desorb excess molecular water from its surface, even at room temperature. Simultaneously, SnO_2 seems to catalyze the dissociation of H_2O and its subsequent chemisorption. This inference is supported by the nearly linear conductance slope within the temperature range of approximately ~ 27 to ~ 160 °C (Figure 17B), evidencing the predominant dissociative conduction mechanism in the presence of humidity for this material.

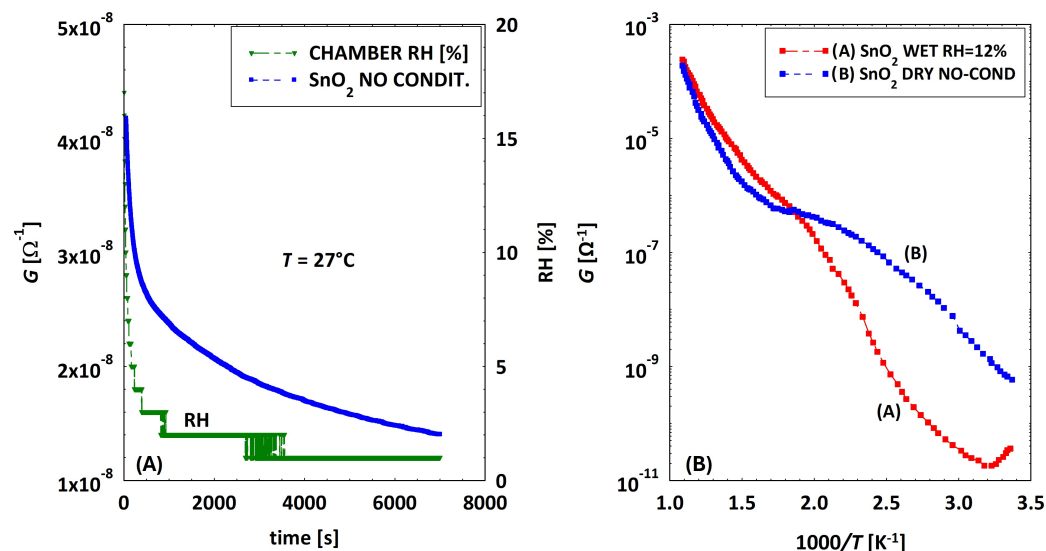


Figure 17. (A) Conductance of a SnO_2 sensor left at $\sim 27^\circ\text{C}$ in dry air flow for two hours. (B) Conductance vs. T for the same SnO_2 sensor in wet ($\text{RH} = 12\%$) atmosphere and in dry air without any preliminary conditioning.

What has been said so far for the pure SnO_2 and the SnTiNbO_2 (remembering that here the Ti content is 30% with respect to Sn) can be further supported introducing some preliminary tests on a slightly different solid Sn–Ti solution with a Ti content of 25%. This particular composition as elucidated in the work by Konstantine R. Hahn et al. [28] is theoretically believed to exhibit the least dissociative interaction with water molecules and is expected to require the lowest water desorption temperature.

This seems to be confirmed by looking at the right panel of Figure 18, in which a conditioned dry air conductance for a $\text{Sn}_{0.75}\text{Ti}_{0.25}\text{O}_2$ (ST25) sensor is compared with conductances in various wet ($\text{RH} = 10, 20, 30, 40$, and 75%) atmospheres. It is clear that for this material, the prevailing water molecular physisorption is repeating as in the case of STN; there also seems to be very little dissociative interaction, and the dry and wet curves overlap at a much lower temperature ($\sim 200^\circ\text{C}$) if compared to that of STN ($\sim 300^\circ\text{C}$) and to that of SnO_2 ($500\text{--}600^\circ\text{C}$). This increased conductivity is a crucial attribute that enhances the material's suitability for sensing applications.

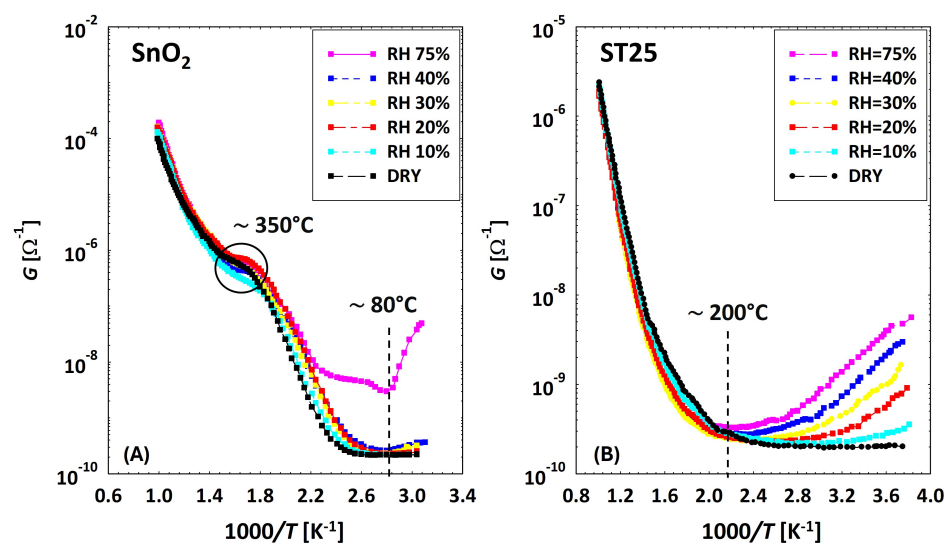


Figure 18. Comparison between conductance shapes for various humidity percentages in function of $1000/T$ for a left (A): SnO_2 sensor, and right (B): $\text{Sn}_{0.75}\text{Ti}_{0.25}\text{O}_2$ (ST25) sensor.

Finally, to complete and confirm the description, the left panel of Figure 18 illustrates how the SnO_2 material appears to inhibit the two molecular water physisorption/conduction mechanisms until higher humidity levels (and the formation of multiple water monolayers) are reached. In the case of SnO_2 , water molecules also begin to dissociate and become chemisorbed at significantly lower temperatures ($\sim 80^\circ\text{C}$) compared to the ST25 material. Notably, the behavior of the plots suggests that, for SnO_2 , water chemisorption occurs in the temperature range of approximately $100\text{--}500^\circ\text{C}$.

In conclusion, it is possible to claim that, even if niobium is added to a binary solid solution of tin and titanium, by looking at the shape of the conductance vs. temperature curves and at the interactions of the material with water, it can be stated that the material maintains the best features of both its constituents: the benefit of the interaction with moisture deriving from titanium oxide, and a good sensitivity towards CO, typical of tin oxide [24], where CO is a standard target gas used to study the sensing properties of materials employed for VOC detection. Another significant benefit of Ni's additional valence over Ti seems to be an increase in electrical conductivity [36].

Finally, the theoretical dissociative or associative nature of the surface–water interactions of pure SnO_2 and pure TiO_2 materials, respectively, appears to be more related to the Ti content in the Sn–Ti solid solutions. Preliminary measurements on a TiO_2 sensor at three different humidity levels show that the associative or dissociative behavior of this material is influenced more by humidity content as seen in Figure 19. Based on what was said about the shapes and slopes of the conductance behavior in function of $1/T$, the $\text{RH} = 20\%$ conductance curve exhibits exclusively dissociative behavior between ~ 180 and $\sim 400^\circ\text{C}$. Similarly, the $\text{RH} = 60\%$ curve also shows dissociative behavior in this range, but the conductance appears additionally modulated by molecular water content between 40 and 100°C .

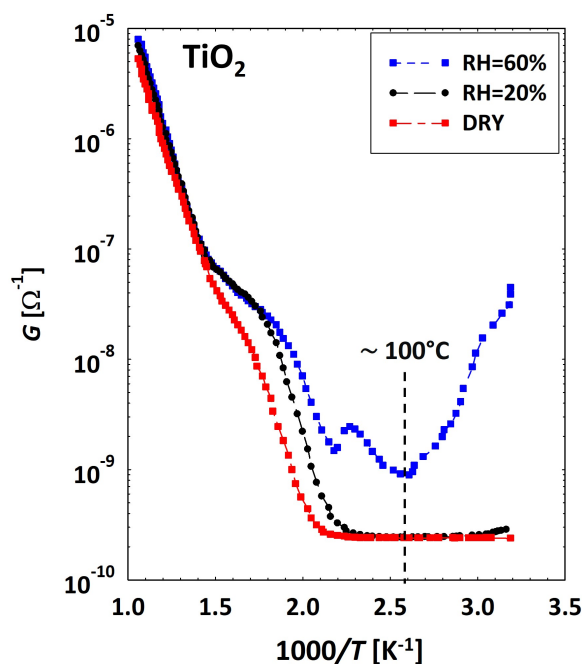


Figure 19. Comparison between conductance shapes for various humidity percentages in function of $1000/T$ for a pure TiO_2 sensor. The dissociative or associative nature of the water–surface interaction here seems to be more water-content related with respect to the more material-related pure SnO_2 and the Sn–Ti mixed solution.

4. Conclusions

SnO_2 was one of the first and most studied materials for gas sensing applications. Its reactivity towards a wide range of gases, especially volatile organic compounds (VOCs), is well known. However, its sensitivity and selectivity are highly temperature dependent,

and moisture is a significant interfering agent, greatly impacting the SnO_2 sensing performance. This study demonstrates that water molecules interact with the tin oxide surface over a wide temperature range, easily reaching temperatures up to 600°C . This is also the temperature range in which SnO_2 exhibits optimal sensing performance for many gases.

The strong dependency on atmospheric conditions is a major limitation for the use of pure SnO_2 in environmental applications and in other contexts where fine control of the surrounding atmosphere is not possible. This necessitates the search for and study of other materials that are less affected by non-target species. Research indicates that pure TiO_2 interacts differently with water molecules and is likely less affected by moisture. However, titanium oxide's higher electrical resistivity makes it less controllable by the simple electronic circuits often used in portable environmental monitoring units.

Promising alternatives are mixtures of SnO_2 and TiO_2 in the form of solid solutions at various percentages [24,34]. This work aimed to investigate and elucidate the molecular processes governing the interactions of water molecules with SnO_2 , TiO_2 , and their mixtures, focusing particularly on the Sn-Ti solid solutions.

The study highlighted the importance of temperature in understanding the conductance kinetics related to sensor surface reactions. Experiments involving heating sensors in both dry and humid conditions across a wide temperature range ($100\text{--}800^\circ\text{C}$) revealed crucial insights into water adsorption/desorption phenomena. Lower temperatures facilitate the removal of physisorbed water, while chemisorbed water requires more energy for desorption. This behavior varies significantly between SnO_2 and SnTiNbO_2 (STN), with STN showing less sensitivity to temperature-induced desorption processes. Starting from theoretical studies suggesting that the crystal structure, unit cell size, and bond character of SnO_2 and TiO_2 play significant roles in their interaction with water, SnO_2 , with its larger unit cell and higher covalent character of Sn-O bonds, promotes water dissociation more effectively than TiO_2 .

Experimental findings corroborate these theoretical insights, indicating that SnO_2 has a higher affinity for moisture adsorption and exhibits distinct dissociative behavior compared to $\text{Sn}_{1-x}\text{Ti}_x\text{O}_2$, which favors molecular adsorption.

The SnO_2 sensor shows significant changes in conductance at low temperatures in wet conditions, highlighting the presence of numerous sites for water chemisorption. In contrast, the STN material shows less pronounced low-temperature interactions with water, indicating a lower activation energy for water chemisorption. Mixed oxides like STN and $\text{Sn}_{75}\text{Ti}_{25}\text{O}_2$ exhibit unique properties that combine the advantages of their constituent oxides. STN retains the sensitivity to carbon monoxide typical of SnO_2 while benefiting from the reduced humidity sensitivity attributed to TiO_2 .

The reduced sensitivity to humidity in STN sensors makes them more reliable in applications where environmental conditions significantly vary. This characteristic is particularly valuable in fields such as environmental monitoring and medical diagnostics, where consistent sensor performance is critical. The study underscores the need for careful material selection and compositional tuning to optimize sensor performance in specific applications. By understanding the interaction mechanisms between water and MOX materials, researchers can design sensors with tailored properties for various environmental conditions.

Further research using this investigative approach on different percentages of tin and titanium in their solid solutions would be significant. This would help lay the foundations for a more precise understanding of water interactions with these materials and allow for the fine-tuning of the tin–titanium ratios in their solid solutions.

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Abbreviations

The following abbreviations are used in this manuscript:

OPAMP	Operational Amplifier
STN	SnTiNbO ₂ tin–titanium–niobium-based metal oxide solid solution
ST25	Sn ₇₅ Ti ₂₅ O ₂ tin–titanium-based solid solution
MOX	Metal Oxides
PW DFT	Plane-Wave Density Functional Theory

Appendix A. General Overview on Arrhenius Plots

The concept at the basis of an Arrhenius diagram derives from the evidence that the chemical reaction rate shows a well-defined dependence with the temperature: the natural logarithm of k (reaction kinetic constant), $\ln(k)$, is linearly proportional to $1/T$, where T is the absolute temperature according to a proportionality coefficient characteristic of the reaction considered. Mathematically, the Arrhenius relationship is expressed by the following Equation (A1):

$$\ln k = \ln a + m \frac{1}{T}, \quad (\text{A1})$$

where $m = -E_a/R$, where a is the pre-exponential factor (an empirical constant which takes into account factors such as the collision frequency between reacting particles, their relative orientation, and the reaction activation entropy), E_a is the *activation energy*, and R is the gas constant. Equation (A1) can be also represented in exponential form as:

$$k = a e^{-E_a/RT}. \quad (\text{A2})$$

Since each law expressed with an exponential temperature dependency can be represented as an Arrhenius diagram, from which the typical activation energy can be easily computed, the conductance of semiconductor oxide thick films, is expressed by the following Arrhenius diagram-like Equation (A3) [9]:

$$G = G_0 e^{-eV_s/k_B T}, \quad (\text{A3})$$

where G is the conductance at the temperature T , G_0 is a constant that takes into account geometrical factors, k_B is the Boltzmann constant, e is the electron charge, and V_s is the barrier height.

The conductance measurements are performed in a wide temperature range (typically from 100 to 800 °C) by increasing the sample temperature to a range of about $\sim 3\text{--}5\text{ }^\circ\text{C} \cdot \text{min}^{-1}$ through a progressive voltage increase. The process is completely automatized, and it produces an experimental curve formed by two or more straight lines, if visualized in a semilog-plot. Each straight line is characterized by a slope, related to a different activation energy of the predominant process (i.e., adsorption or desorption of chemical species at the semiconductor surface) in a certain temperature range.

It is worth noting that the linearization of the exponential term $\exp(-eV_s/k_B T)$ derives from the natural logarithm of the ratio G/G_0 , which is a dimensionless quantity:

$$\ln \frac{G}{G_0} = -\frac{eV_S}{k_B T}, \quad (\text{A4})$$

and since the constant G_0 can be calculated only via the determination of the potential barrier at a fixed temperature [37] (procedure not performed for this work purposes), the behavior of the conductance G , (omitting information on the potential barrier) can be shown anyway in function of $1/T$, by plotting it in a log ordinate axis. To improve the paper readability, this plot will be, in the whole document, called interchangeably as an “Arrhenius-like” plot or “Arrhenius*” plot.

Appendix B. Signal Reading: The Operational Amplifier (OPAMP)

The state of the art in gas sensor signal reading foresees the use of an operational amplifier (OPAMP) connected in inverting configuration (Figure A1), providing a voltage output (V_{out}), directly proportional to the sensor conductivity.

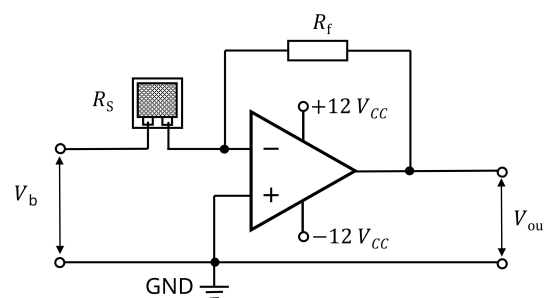


Figure A1. Sensor film electrical acquisition circuit; the operational amplifier is in an inverting configuration.

Indeed, V_{out} is proportional to the ratio R_f/R_S , where R_f and R_S are the feedback resistor and the sensor film, respectively, and the bias voltage V_b is fixed to -5 V. Since the chemical interactions between the gaseous atmosphere and the sensor sensitive layer induce its conductivity variation, the OPAMP-based transduction circuit produces an analog voltage output that is then transformed in a digital one by means of an analog-to-digital converter. The transfer function of the circuit is:

$$V_{out} = -\frac{R_f}{R_S} V_b, \quad (\text{A5})$$

so the sensor resistance is:

$$R_S = -V_b \frac{R_f}{V_{out}}. \quad (\text{A6})$$

This layout ensures both the linearity of the transfer function Equation (A5), and the constant polarization of the sensor film at the fixed voltage (-5 V), and it is capable of reading signals in the range 0 to 12 V. It also enhances the resolution on the signal variations and reduces the noise at low currents by means of ad hoc filtering.

Appendix B.1. Uncertainties on Film Conductance Measurements

It is possible to quantify the uncertainties associated with the absolute conductance values of the sensor film. As seen in Appendix A, Arrhenius plots, particularly in this study, where we analyzed conductance variations as a function of the inverse temperature, play a significant role. Therefore, it becomes crucial to assess the individual contributions of each component within the electronic circuit to the overall results, and how this evaluation impacts the final outcome. In Appendix B, Equation (A5) can be used to calculate the conductance of the sensor film G_S :

$$G_S = \frac{V_{out}}{V_b R_f}, \quad (\text{A7})$$

where all the quantities have already been mentioned. So, the uncertainty on G is calculated by the usual uncertainty propagation formula:

$$\Delta G = \sqrt{\left(\frac{\partial G}{\partial V_{\text{out}}} \Delta V_{\text{out}}\right)^2 + \left(\frac{\partial G}{\partial V_b} \Delta V_b\right)^2 + \left(\frac{\partial G}{\partial R_f} \Delta R_f\right)^2}, \quad (\text{A8})$$

with:

$$\begin{aligned} \frac{\partial G}{\partial V_{\text{out}}} &= \frac{1}{V_b R_f}, \\ \frac{\partial G}{\partial V_b} &= -\frac{V_{\text{out}}}{R_f V_b^2}, \\ \frac{\partial G}{\partial R_f} &= -\frac{V_{\text{out}}}{V_b R_f^2}. \end{aligned} \quad (\text{A9})$$

with a given voltage V_b of -5 ± 0.25 V, an uncertainty of 5% on R_f , and an uncertainty on V_{out} as read by the *Keithley 2000* laboratory multimeter, whose accuracy is given by the instrument manual to be:

$$\pm(30 \text{ ppm of reading} + 5 \text{ ppm of range}), \quad (\text{A10})$$

this value is, for simplicity, rounded to the third decimal digit, so the final uncertainty on V_{out} becomes $\pm 0.5 \times 10^{-3}$ V. The total uncertainty for each measured value of conductance G can be computed. To illustrate this, in the Arrhenius-like plot chart depicted in Figure A2, each conductance value is accompanied by its corresponding vertical uncertainty bars. $\Delta G/G$ results in fluctuating slightly around 5% over the entire range (ruled by the greater uncertainty relative to the feedback resistances R_f).

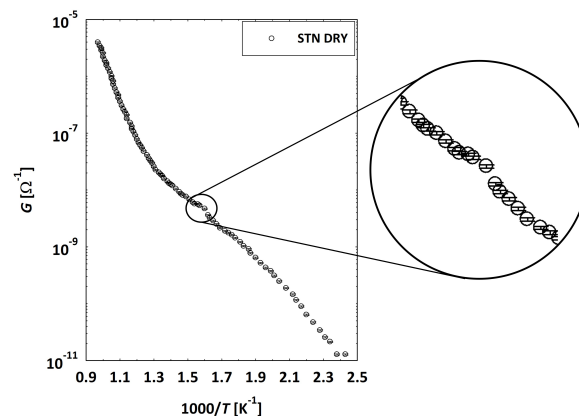


Figure A2. Illustrative scatter plot of conduction trend in function of inverse temperature, aiming to show how uncertainties on G propagate over a wide range of values. Vertical uncertainty bars are barely visible inside the circles centered on the nominal values.

Due to the fact that error bars are barely visible on this type of graph, this value of 5% should be taken as the maximum uncertainty on all the Arrhenius-like plots in the whole document (please note that the calculated uncertainties provided in this appendix should be considered inherent to all charts presented in this work; while they are not explicitly displayed on the charts due to their minimal impact, they are integral to the overall data interpretation and analysis).

Appendix C. Sensor Temperature Regulation

The term “meander” is commonly used in electronics to describe a specific type of resistor layout or pattern. It typically refers to a resistor that is arranged in a serpentine

or zigzag pattern, resembling the shape of a meandering river. This layout increases the overall length of the resistor within a limited space, allowing to achieve a greater surface area and higher resistance values. The alumina supports are equipped with a screen-printed platinum meander acting as a heater by the Joule effect (Figure A3).

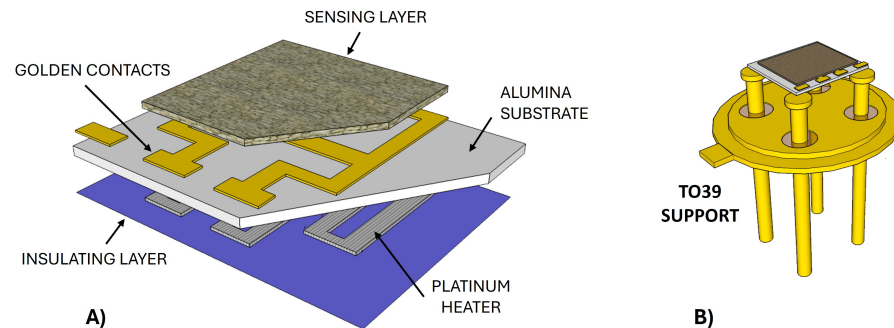


Figure A3. Left (A): 3D cutaway drawing of the various layers that make up a sensor with its alumina support. Right (B): final configuration of a sensor bonded on the TO39 support housing.

Their typical ohmic resistance values, at ambient temperature, are (by custom request) randomly distributed in the range of about 8 and 13 Ω . By applying a voltage in the range 1–7 V, these resistors, and so the surrounding alumina substrate equipped with the metal oxide film, can heat up in a temperature range of about 100–800 $^{\circ}\text{C}$.

The Callendar–Van Dusen Relation

As commonly used for commercial applications of RTD thermometers, the relationship between the ohmic resistance R_T and temperature T is given by the **Callendar–Van Dusen** formula (this two-parameter formula is valid as long as $T \geq 0$ $^{\circ}\text{C}$):

$$R_T = R_0(1 + \alpha T + \beta T^2), \quad (\text{A11})$$

where R_0 is the ohmic resistance at 0 $^{\circ}\text{C}$, and α [$^{\circ}\text{C}^{-1}$] and β [$^{\circ}\text{C}^{-2}$] are the experimentally determined parameters. Since every heater has its own R_0 value and sensors cannot be easily brought to 0 $^{\circ}\text{C}$, this parameter has to be calculated by extrapolating it from Equation (A11). The procedure consists in positioning the sensor in a closed box at room temperature and measuring together T and R_T . The uncertainty on R_0 can be estimated with the propagation Formula (A12):

$$\Delta R_0 = \sqrt{\left(\frac{\partial R_0}{\partial R} \cdot \Delta R\right)^2 + \left(\frac{\partial R_0}{\partial T} \cdot \Delta T\right)^2 + \left(\frac{\partial R_0}{\partial \alpha} \cdot s_{\alpha_{\text{best}}}\right)^2 + \left(\frac{\partial R_0}{\partial \beta} \cdot s_{\beta_{\text{best}}}\right)^2}, \quad (\text{A12})$$

where $s_{\alpha_{\text{best}}}$ and $s_{\beta_{\text{best}}}$ are the α and β standard deviations, respectively. The previously calculated uncertainties on α and β , along with the uncertainties on T (estimated to be of 0.2% at 25 $^{\circ}\text{C}$) and on R (estimated to be of the order of 0.01%) give rise to a percent uncertainty of $\sim 1\%$ on R_0 .

By supplying a voltage to the heater and measuring the corresponding flowing current, sensors in the test chamber can be brought up to operating temperature. This allows for the calculation of the heater's actual resistance (with a cumulative uncertainty of $\sim 1\%$). In this way, modulating V , the heater can be set at the theoretical calculated value R_T corresponding to the desired temperature.

In temperature stimulated analysis, like, for example, Arrhenius plots, the sweep in temperature is forced by varying the heater voltage, generally from 1 to 7 V, and the temperature is calculated a posteriori by the inverse Equation (A13):

$$T = \frac{-\alpha R_0 + \sqrt{a^2 R_0^2 - 4(R_0 - R)\beta R_0}}{2\beta R_0}. \quad (\text{A13})$$

Similarly to Equation (A12), uncertainty propagation can be calculated for T ; it strongly depends on the value of the measured R_T and on its uncertainty as can be deduced from the charts of Figure A4A,B.

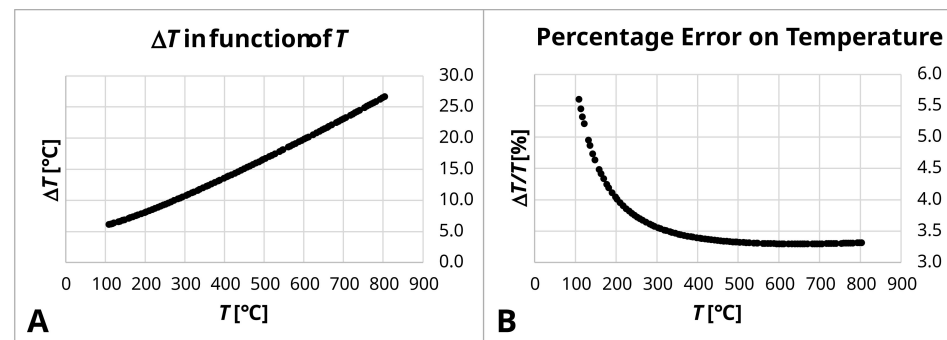


Figure A4. (A) Absolute uncertainty on the temperature as a function of T itself. (B) Percentage uncertainty on the temperature as a function of T itself.

In this way, by fitting the data in Figure A4A with a power law of the type $y = a + bx^c$, uncertainties on temperature can be given for every T . With the calculated a , b , and c values, the equation becomes:

$$\Delta T = 4.134 + 0.0053 T^{1.24}. \quad (\text{A14})$$

Following this rule, uncertainties on T spans from ~6% to 3% over the range from low to high (again, as for the uncertainties on the conductance G , error bars for the temperature T are not shown on any of the document charts; a visual reference for their magnitude is shown in Figure A5 once and for all). To complete the topic, an example of conductance in the function of inverse temperature for a generic $\text{Sn}_{0.7}\text{Ti}_{0.3}\text{O}_2$ (STN) is plotted in Figure A5. Uncertainties on the abscissa (that span gradually from 1 to 3% starting from right to left), which were obtained using the previously calculated ΔT values according to Equation (A14), have been incorporated into the computation of $1000/T$, converting T to Kelvin.

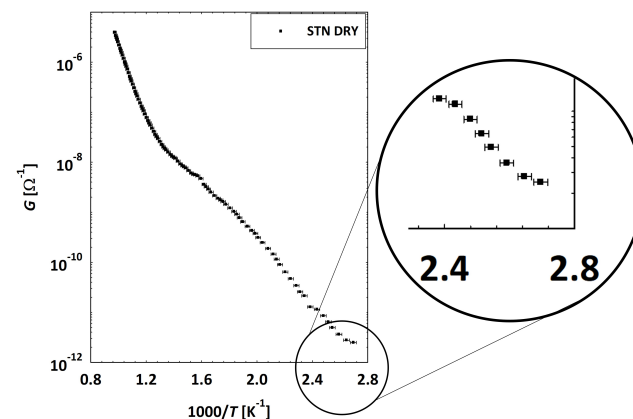


Figure A5. An illustrative scatter plot depicting conductance as a function of inverse temperature, demonstrating the propagation of uncertainties in temperature (T) across a wide temperature range (100–800 °C).

Appendix D. Water–Surface Interactions

The adsorption of water vapor from the surrounding atmosphere can be very marked on the sensor surface, and the presence of humidity can, in many cases, inhibit the sensors' response to other gases. Moreover, water vapor also requires high temperatures to be removed from the sensor film. The full understanding of the cross-sensitivity mechanism in the presence of humidity for many metal oxides sensors is still in progress, although for the most widely employed materials (e.g., SnO₂), two main processes can be described after the dissociation of an H₂O molecule in an H atom and an OH group Equation (A15):



and the resultant ions are adsorbed onto the metal oxide surface. The first mechanism [38] is based on the hypothesis that if a water molecule interacts with two metal oxides, then two Sn–OH dipoles are formed from the dissociated hydroxyl ion and by a combination of a hydrogen ion and a lattice oxygen (see Figure A6A) [39].

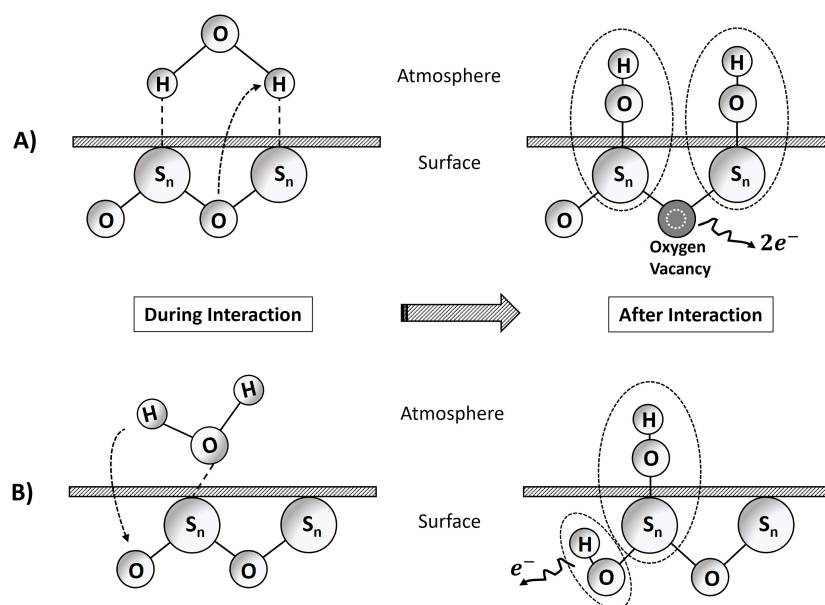
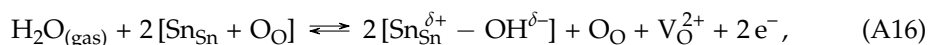
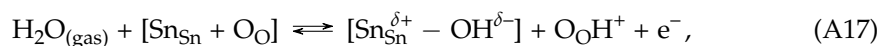


Figure A6. Mechanism of humidity adsorption on the semiconductor oxide surface; (A) one water molecule for two metal sites, and (B) one water molecule per one metal site.

Then, two free electrons are donated to the conduction band from each oxygen vacancy (V_O²⁺) according to Equation (A16):



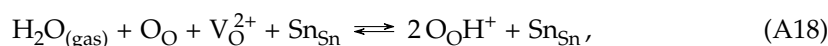
where (Sn_{Sn} – OH) refers to an isolated (terminal) hydroxyl or OH group (dipole). The notation here partially follows the Kröger–Vink convention, where lattice sites occupied by a particular species are denoted as subscripts. For instance, O_O represents an oxygen atom in an oxygen lattice position. The second process (see Figure A6B) involves a reaction between one water molecule and a single metal oxide, the dissociated hydroxyl ions are bonded with each metal site, and the hydrogen ion diffuses into the lattice to bind with the lattice oxygen, thus forming two hydroxyl bonds per metal site. The process is governed by the reaction Equation (A17):



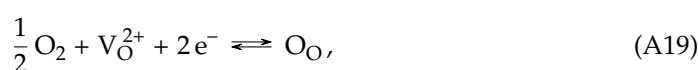
where O_OH⁺ is the rooted species (lattice). The reaction implies the homolytic dissociation of water and the reaction of the neutral H atom with the lattice oxygen. Having a lower

affinity to electrons, the built-up rooted OH group can be ionized, turning into a donor (releasing an electron to conduction band).

A third mechanism [40] involves oxygen vacancies instead of Sn sites as one of the most common dissociative acid/base reactions on metal oxide surfaces: the deprotonation of an adsorbate to produce hydroxyl groups. According to the predominant direction of charge transfer, adsorbates can be categorized as electron *donors* or *acceptors*, but the involved electrons are never in any sense “free”. For water adsorption (see Figure A7), this reaction is Equation (A18):



where Sn_{Sn} is only a spectator species. In this case, the reducing effect of water may not only be related to the dissociative adsorption according to Equations (A16) and (A17), but associatively adsorbed water molecules may directly act as an electron donor, or obstruct the ionosorption of oxygen (Equation (A19)):



e.g., by competing for the same surface sites, which are involved in oxygen ionosorption. Equation (A19) can be read for both sides, in order to take an oxygen from a lattice site inside the crystal and land it on the surface, and vice versa. This phenomenon can also be called “in/out diffusion”.

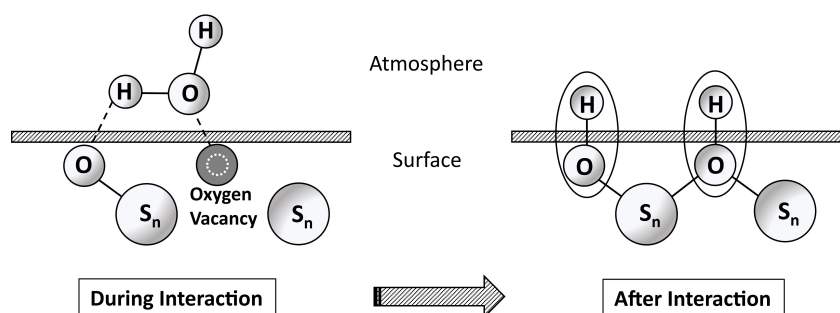


Figure A7. Deprotonation of a water molecule to form hydroxyl groups. In this case, the adsorbate can act as an electron donor or obstruct the ionosorption of oxygen, competing for the same surface state.

In line with another proposed interaction [25], electron injection does not occur directly from the water molecule; instead, it arises from adjacent adsorbed oxygen ions. When a polar water molecule becomes adsorbed next to an oxygen ion, this interaction leads to changes in the water molecule’s energy level and the rate at which electrons are injected and extracted. Consequently, this process results in a reduction in the density of adsorbed negatively charged oxygen. This event was also predicted to be plausible, especially in SnO_2 , by Andrei V. Bandura et al. using plane-wave density functional theory (PW DFT) simulations. The protonation of one bridging oxygen on the (110) SnO_2 surface through dissociation leads to a reduction in the basicity of the adjoining bare bridging oxygen, and in general a reorganization of the electronic states after the adsorption of H_2O molecules [11].

Furthermore, another mechanism [41] must be taken into account: the co-adsorption of water molecules onto another adsorbate layer. In these studies, it was hypothesized that there would be a formation of water vapor layers on the surface, due to high humidity levels. At low humidity levels, the adsorption of water molecules on the metal oxide surface obeys the chemisorption process [42] as explained by Equations (A16)–(A18), whereas at higher humidity levels, all the active sites on the metal oxide surface are occupied by water molecules. In addition, another water molecule layer can be formed following the physisorption process. Resistance sharply decreases during chemisorption, whereas it is slowed down during physisorption in a higher humidity range, and then the increase in

conductivity slows down quickly during co-adsorption at higher humidity levels. At this point, the conduction can occur by the Grotthuss transport mechanism, said to happen at low temperatures ($\sim 150\text{--}300\text{ }^{\circ}\text{C}$) as the humidity percentage increases (until 60% [43]), also known as the proton hopping mechanism (see Figure A8).

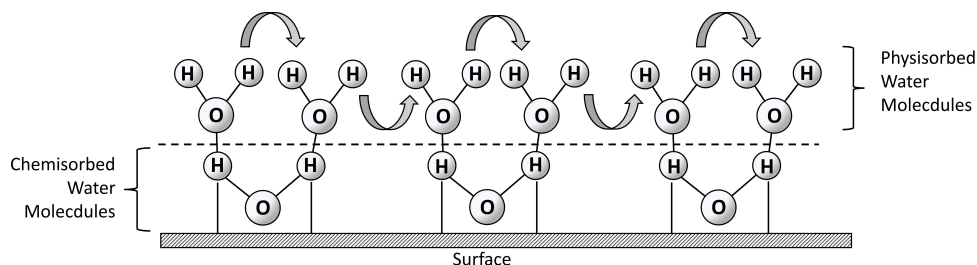


Figure A8. Schematic representation of humidity sensing in the presence of a physisorbed layer over a chemisorbed one in high atmospheric humidity conditions. The proton hopping mechanism is represented by curved arrows.

At a greater humidity percentage (above 80%; as can be seen in this work, percentages can vary substantially depending on materials, temperatures, surface geometry, and also boundary and initial atmospheric conditions), the mechanism is more likely to be associated to hydronium (H_3O^+) transport (see Figure A9 [43]).

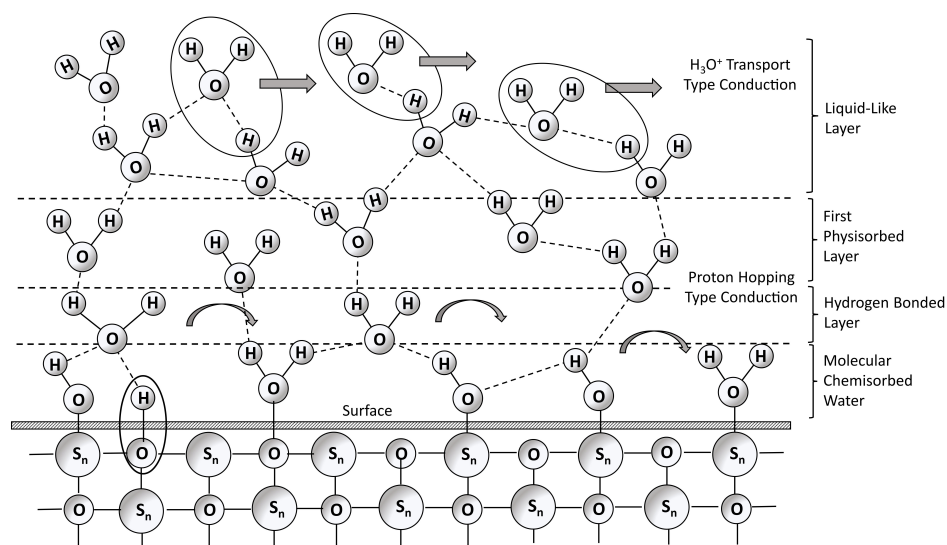


Figure A9. Schematic representation of conductivity governed by the H_3O^+ transport mechanism in the presence of a “liquid-like” substrate over an “ice-like” substrate and a hydroxylated chemisorbed substrate.

Thus, it can be concluded that (in this regime of low-temperature and high-humidity ranges) the conduction mechanism changes from Grotthuss to a vehicle-type mechanism when the water layer changes from “ice-like” to “liquid-like” in its properties [43]. The minimum required temperature to completely dehydroxylate a semiconductor oxide surface is not a priori determined; therefore, the hypothesis is that a temperature of at least $500\text{--}600\text{ }^{\circ}\text{C}$ can be sufficient. However, as explained throughout this work, a temperature of around $700\text{ }^{\circ}\text{C}$ for about 10 min in dry air can provide certainty about the process.

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